MPG No. 6.3.19 2′2′2 (2’2’2 setting) [Type III, orthorhombic] [M tensor]

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} 0 & 0 & M_z \end{bmatrix}$$

$$M_z = \mathbb{M}_z^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} 0 & M_{xy} & 0 \\ M_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xy} = \mathbb{M}_{xy}^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & M_{xy} & 0 \\ -M_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xy} = \mathbb{T}_z^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & M_{xxz} \\ 0 & 0 & M_{yyz} \\ 0 & 0 & M_{zzz} \\ 0 & M_{yyz} & 0 \\ M_{xxz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxz} = \mathbb{M}_z^{(1)} - \mathbb{M}_z^{\alpha(1)} + \mathbb{M}_z^{\beta(1)}$$
$$M_{yyz} = \mathbb{M}_z^{(1)} - \mathbb{M}_z^{\alpha(1)} - \mathbb{M}_z^{\beta(1)}$$
$$M_{zzz} = 3\mathbb{M}_z^{(1)} + 2\mathbb{M}_z^{\alpha(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & M_{xxz} \\ 0 & 0 & M_{yyz} \\ 0 & 0 & 0 \\ 0 & -\frac{M_{yyz}}{2} & 0 \\ -\frac{M_{xxz}}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxz} = 2\mathbb{M}_z^{(2)} + 2\mathbb{T}_{xy}^{(1)}$$
$$M_{yyz} = 2\mathbb{M}_z^{(2)} - 2\mathbb{T}_{xy}^{(1)}$$
$$\frac{M_{yyz}}{2} = \mathbb{M}_z^{(2)} - \mathbb{T}_{xy}^{(1)}$$
$$\frac{M_{xxz}}{2} = \mathbb{M}_z^{(2)} + \mathbb{T}_{xy}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & M_{yzy} & 0 \\ M_{zxx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzy} = \mathbb{M}_{z}^{(3)} + \mathbb{T}_{xy}^{(2)}$$
$$M_{zxx} = -\mathbb{M}_{z}^{(3)} + \mathbb{T}_{xy}^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & M_{yyxy} \\ 0 & 0 & 0 & 0 & 0 & M_{zzxy} \\ 0 & 0 & 0 & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & M_{yzzx} & 0 & 0 \\ M_{xxxy} & M_{yyxy} & M_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxy} = \mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)}$$
$$M_{yyxy} = -\mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)}$$
$$M_{zzxy} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)}$$
$$M_{yzzx} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & M_{yyxy} \\ 0 & 0 & 0 & 0 & 0 & M_{zzxy} \\ 0 & 0 & 0 & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & -M_{yzzx} & 0 & 0 \\ -M_{xxxy} & -M_{yyxy} & -M_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxy} = \mathbb{M}_{xy}^{(3)} + 2\mathbb{T}_{z}^{(1)} - 2\mathbb{T}_{z}^{\alpha(1)} + 2\mathbb{T}_{z}^{\beta(1)}$$
$$M_{yyxy} = \mathbb{M}_{xy}^{(3)} - 2\mathbb{T}_{z}^{(1)} + 2\mathbb{T}_{z}^{\alpha(1)} + 2\mathbb{T}_{z}^{\beta(1)}$$
$$M_{zzxy} = \mathbb{M}_{xy}^{(3)} - 4\mathbb{T}_{z}^{\beta(1)}$$
$$M_{yzzx} = -\mathbb{T}_{z}^{(1)} - 4\mathbb{T}_{z}^{\alpha(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & M_{yzzx} & 0 \\ M_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = 2\mathbb{M}_{xy}^{(4)}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & M_{yzzx} & 0 \\ -M_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = -\mathbb{T}_{z}^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & M_{xxxy} \\ 0 & 0 & M_{yyxy} \\ 0 & 0 & M_{zzxy} \\ 0 & M_{yzzx} & 0 \\ M_{zxyz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxy} = -2\mathbb{M}_{xy}^{(5)} + \mathbb{T}_z^{(3)} - \mathbb{T}_z^{\alpha(2)} + \mathbb{T}_z^{\beta(2)}$$
$$M_{yyxy} = 2\mathbb{M}_{xy}^{(5)} + \mathbb{T}_z^{(3)} - \mathbb{T}_z^{\alpha(2)} - \mathbb{T}_z^{\beta(2)}$$
$$M_{zzxy} = \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{(4)} + 2\mathbb{T}_z^{\alpha(2)}$$
$$M_{yzzx} = -\mathbb{M}_{xy}^{(5)} + \mathbb{T}_z^{(4)} - \mathbb{T}_z^{\alpha(2)} - \mathbb{T}_z^{\beta(2)}$$
$$M_{zxyz} = \mathbb{M}_{xy}^{(5)} + \mathbb{T}_z^{(4)} - \mathbb{T}_z^{\alpha(2)} + \mathbb{T}_z^{\beta(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & M_{zxyz} & 0 & 0 \\ M_{xyxx} & M_{xyyy} & M_{xyzz} & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = \mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} + \mathbb{T}_z^{\beta(3)}$$
$$M_{zxyz} = -\mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} - \mathbb{T}_z^{\beta(3)}$$
$$M_{xyxx} = -2\mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)} + \mathbb{T}_z^{\beta(3)}$$
$$M_{xyyy} = 2\mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)} - \mathbb{T}_z^{\beta(3)}$$
$$M_{xyzz} = \mathbb{T}_z^{(5)} + 2\mathbb{T}_z^{(6)} + 2\mathbb{T}_z^{\alpha(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{xxxy} & 0 & 0 & M_{xxyx} \\ 0 & 0 & 0 & 0 & 0 & M_{yyxy} & 0 & 0 & M_{yyyx} \\ 0 & 0 & 0 & 0 & 0 & M_{zzxy} & 0 & 0 & M_{zzyx} \\ 0 & 0 & 0 & 0 & M_{yzzx} & 0 & 0 & M_{yzxz} & 0 \\ 0 & 0 & 0 & M_{zxyz} & 0 & 0 & M_{zxzy} & 0 & 0 \\ M_{xyxx} & M_{xyyy} & M_{xyzz} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxy} = \mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)} + 2\mathbb{T}_z^{(1)} + \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} + 2\mathbb{T}_z^{\beta(1)} + \mathbb{T}_z^{\beta(2)}$$
$$M_{xxyx} = \mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_{xy}^{(5)} + 2\mathbb{T}_z^{(1)} - \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)} + 2\mathbb{T}_z^{\beta(1)} - \mathbb{T}_z^{\beta(2)}$$
$$M_{yyxy} = -\mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_{xy}^{(5)} - 2\mathbb{T}_z^{(1)} + \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} + 2\mathbb{T}_z^{\beta(1)} - \mathbb{T}_z^{\beta(2)}$$
$$M_{yyyx} = -\mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)} - 2\mathbb{T}_z^{(1)} - \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)} + 2\mathbb{T}_z^{\beta(1)} + \mathbb{T}_z^{\beta(2)}$$
$$M_{zzxy} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + \mathbb{M}_{xy}^{(3)} + \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{(4)} + 2\mathbb{T}_z^{\alpha(2)} - 4\mathbb{T}_z^{\beta(1)}$$
$$M_{zzyx} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + \mathbb{M}_{xy}^{(3)} - \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{(4)} - 2\mathbb{T}_z^{\alpha(2)} - 4\mathbb{T}_z^{\beta(1)}$$
$$M_{yzzx} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(5)} - \mathbb{T}_z^{(1)} + \mathbb{T}_z^{(4)} - 4\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} - \mathbb{T}_z^{\beta(2)}$$
$$M_{yzxz} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(5)} - \mathbb{T}_z^{(1)} - \mathbb{T}_z^{(4)} - 4\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)} + \mathbb{T}_z^{\beta(2)}$$
$$M_{zxyz} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(5)} + \mathbb{T}_z^{(1)} + \mathbb{T}_z^{(4)} + 4\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} + \mathbb{T}_z^{\beta(2)}$$
$$M_{zxzy} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(5)} + \mathbb{T}_z^{(1)} - \mathbb{T}_z^{(4)} + 4\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)} - \mathbb{T}_z^{\beta(2)}$$
$$M_{xyxx} = \mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(3)} - 2\mathbb{T}_z^{(1)} + 2\mathbb{T}_z^{\alpha(1)} - 2\mathbb{T}_z^{\beta(1)}$$
$$M_{xyyy} = -\mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(3)} + 2\mathbb{T}_z^{(1)} - 2\mathbb{T}_z^{\alpha(1)} - 2\mathbb{T}_z^{\beta(1)}$$
$$M_{xyzz} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} - \mathbb{M}_{xy}^{(3)} + 4\mathbb{T}_z^{\beta(1)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{yzzx} & 0 & 0 & M_{yzxz} & 0 \\ 0 & 0 & 0 & M_{zxyz} & 0 & 0 & M_{zxzy} & 0 & 0 \\ M_{xyxx} & M_{xyyy} & M_{xyzz} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$
\begin{aligned}
M_{yzzx} &= 2\mathbb{M}_{xy}^{(4)} + \mathbb{M}_{xy}^{(6)} - \mathbb{T}_{z}^{(2)} + \mathbb{T}_{z}^{(6)} - \mathbb{T}_{z}^{\alpha(3)} + \mathbb{T}_{z}^{\beta(3)} \\
M_{yzxz} &= -2\mathbb{M}_{xy}^{(4)} + \mathbb{M}_{xy}^{(6)} + \mathbb{T}_{z}^{(2)} + \mathbb{T}_{z}^{(6)} - \mathbb{T}_{z}^{\alpha(3)} + \mathbb{T}_{z}^{\beta(3)} \\
M_{zxyz} &= 2\mathbb{M}_{xy}^{(4)} - \mathbb{M}_{xy}^{(6)} + \mathbb{T}_{z}^{(2)} + \mathbb{T}_{z}^{(6)} - \mathbb{T}_{z}^{\alpha(3)} - \mathbb{T}_{z}^{\beta(3)} \\
M_{zxzy} &= -2\mathbb{M}_{xy}^{(4)} - \mathbb{M}_{xy}^{(6)} - \mathbb{T}_{z}^{(2)} + \mathbb{T}_{z}^{(6)} - \mathbb{T}_{z}^{\alpha(3)} - \mathbb{T}_{z}^{\beta(3)} \\
M_{xyxx} &= -2\mathbb{M}_{xy}^{(6)} + \mathbb{T}_{z}^{(5)} - \mathbb{T}_{z}^{\alpha(3)} + \mathbb{T}_{z}^{\beta(3)} \\
M_{xyyy} &= 2\mathbb{M}_{xy}^{(6)} + \mathbb{T}_{z}^{(5)} - \mathbb{T}_{z}^{\alpha(3)} - \mathbb{T}_{z}^{\beta(3)} \\
M_{xyzz} &= \mathbb{T}_{z}^{(5)} + 2\mathbb{T}_{z}^{(6)} + 2\mathbb{T}_{z}^{\alpha(3)}
\end{aligned}
$$

* Rank 4 tensor (t).

$$
\begin{bmatrix}
0 & M_{xxxy} & 0 \\
M_{yyyx} & 0 & 0 \\
0 & 0 & 0 \\
0 & 0 & 0 \\
0 & M_{zzxy} & 0 \\
M_{xxyx} & 0 & 0 \\
M_{yzzx} & 0 & 0 \\
0 & 0 & 0 \\
0 & M_{xyyy} & 0 \\
0 & 0 & \frac{M_{xxxy}}{2} - \frac{M_{xxyx}}{2} + M_{xyyy} - M_{yyyx} + M_{zzxy}
\end{bmatrix}
$$

$$
\begin{aligned}
M_{xxxy} &= \mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)} \\
M_{yyyx} &= -\mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)} \\
M_{zzxy} &= 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + \mathbb{M}_{xy}^{(3)} \\
M_{xxyx} &= \mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_{xy}^{(5)} \\
M_{yzzx} &= 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(5)} \\
M_{xyyy} &= -\mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(3)}
\end{aligned}
$$